

Instawards Statement of Work (SOW)

30-Day Scoped Engagement

1. Project & Team Information

Field	Value
Project Name	QWATT (formerly WattCoin) — a DeFi energy meter
Builder / Team	Jae Hee (Jay) Kim, Diego Ostertag
Primary Contact	Jae Hee (Jay) Kim — jay@quorelia.org
Ambassador Chapter	Chile
Ambassador Chapter Lead	Bastian Koh
Date Submitted	September 2026
Suggested Sprint Start Date	September 14, 2026
Duration	30 days (4-week sprint)

2. Instawards Overview & Intent

2.1 Purpose

QWATT, formerly WattCoin, is a DeFi energy meter: a verification layer that turns metered renewable generation into independently verifiable records on Stellar. This Instaward focuses on delivering a meter-agnostic verification engine for that layer, with multi-party authorization on Stellar Testnet.

Rather than building around one specific device, this sprint delivers a software layer that accepts the output of any standard energy meter — Modbus, HTTP/JSON or CSV — normalizes it to one record, runs it through open, versioned physics gates, and anchors the accepted evidence on Stellar through a 2-of-3 multisig in which an independent verifier re-runs every check from published data.

The value this creates is simple: the cost of trusting a small generator's claim falls to almost nothing. Today a building that shares one rooftop array, a client billed under a behind-the-meter PPA, or a registry issuing renewable attributes can only trust a small installation by sending a person or taking the owner's word. A record anyone can check in a browser makes thousands of small roofs claimable and auditable as a class. The token stays what it is: a receipt for one verified megawatt-hour, minted only by physics and multi-party agreement, and consumed only when someone retires it to make a true claim.

The result will be a fully demonstrable pilot: seven consecutive days of unattended operation on Stellar Testnet, where a reviewer with no access to the team can take any anchored batch, download its evidence, re-run the gates and confirm the on-chain hash — and where no single key, including the team's, can write a record on its own.

This is a resubmission. Compared with the previous proposal it removes all hardware, the offtake agreement and token issuance from scope, and replaces single-operator anchoring with multi-party authorization.

3. Problem Statement & Objective

Problem Being Addressed

QWATT already has the technical foundation for verifying solar generation: a versioned gate specification, a shared verification engine with a 29-case conformance suite, seven adversarial validators with 24 tests, and a live testnet asset. What it lacks is the path from a real meter to a record that someone outside the team can trust.

Today there is no integrated flow allowing:

- a standard meter of any brand to feed the engine without custom code
- an interval to be assembled into a hash-chained, publicly downloadable evidence batch
- an anchoring transaction to require a signature from a party the team does not control
- an outside reviewer to reconcile what is on-chain against what was published
- a rejected interval to be visible to anyone at all

Without this, QWATT cannot demonstrate that Stellar can carry independently verifiable energy records, or validate the verification model with real data.

Objective

Within 30 days QWATT will deliver a reviewer-verifiable, meter-agnostic verification pipeline on Stellar Testnet.

Any Modbus, HTTP or CSV meter source will be able to feed the engine through configuration alone; every batch — accepted or rejected — will be published with its hash and reason; rejected batches are never anchored; accepted batches are anchored on Stellar only when a proposer, an independent verifier and an off-team third signer agree; and the pipeline will run unattended for the final seven days of the sprint, with a public reconciliation page that lets anyone confirm the chain end-to-end from Horizon and the evidence feed.

4. Scope of Work

Deliverable 1 (Week 1)

Meter-Agnostic Ingestion

Build:

- Modbus RTU/TCP adapter with the register map as configuration (JSON), not code
- HTTP/JSON push adapter
- CSV / file replay adapter
- Normalization to the versioned interval schema (interval.schema.json)
- Two reference register maps: PZEM-017 and EPEver Tracer — the Phase 1 bench instruments
- Raw-payload hashing so the normalization itself is auditable
- Adapter test harness: the same input through all three adapters yields byte-identical output

Why this matters

This is what makes the engine usable by any renewable installation: a new meter becomes a configuration file, and the verification logic never changes to accommodate hardware.

Deliverable 2 (Week 2)

Evidence Batches & Hash Chain

Build:

- Batch definition (batch.schema.json): batch id, start and end timestamps, interval count, accepted and rejected interval counts, schema version, gate specification version, previous batch hash, Merkle root over raw samples, cumulative energy, and the SHA-256 of the canonical JSON
- Batch assembly: intervals → canonical JSON → SHA-256 → prev_hash chain; energies stored as integers so no floating-point value ever reaches a hash
- Publication and anchoring are distinct: every batch, accepted or rejected, is published; only accepted batches are anchored on Stellar
- Accepted and rejected ledgers with reason codes (append-only JSONL)
- Evidence manifest format for public download
- Batch-level conformance cases added to the existing suite
- Scale test: one month of a public PV dataset (historical data — the replay runs in minutes) pushed through the batch pipeline to exercise chaining across thousands of intervals

Why this matters

A verifier that shows only its successes has demonstrated nothing. The chain makes every record tamper-evident, and the rejection ledger makes the checks falsifiable by anyone.

Deliverable 3 (Week 3)

Multi-Party Authorization on Stellar

Build:

- Anchoring account configured with native Stellar multisig, medium threshold 2-of-3
- Proposer service: signs batches the engine accepted
- Independent verifier service on a separate host with a separate key: downloads the published batch, recomputes the hash, re-runs the gates and signs only on agreement — it never receives data from the proposer directly
- Third signer key held off-team, designated at kickoff
- One atomic transaction per batch: manageData evidence hash + manageData cumulative total + payment op, using the existing testnet asset as the anchoring vehicle only

- Four published test transactions: (A) proposer signs alone — rejected by the network; (B) verifier signs alone — rejected; (C) proposer and verifier sign — anchored; (D) evidence altered after the verifier's read — the verifier refuses and nothing anchors
- Independence acceptance test: the verifier's only input path is the public evidence feed — a batch the proposer holds but has not published cannot be signed
- Live run starts as soon as the first multisig batch anchors (target day 21, latest day 23)

Why this matters

Enforcement moves from policy to the ledger. No single key — including the team's — can write a record, and the party that must agree re-derives its decision from public data.

Deliverable 4 (Week 4)

Public Evidence Feed, Reconciliation & Seven-Day Live Run

Build:

- Public evidence feed: every batch, accepted and rejected, with hash and reason — published, while anchoring stays accepted-only
- Browser reconciliation page reading Horizon: anchored hashes ↔ published batches, cumulative totals, sequence gaps
- Recovery logic in place before day 21: restart-safe state, Horizon timeouts and transaction retries, duplicate-interval rejection, clock checks, and safe behaviour when the verifier or a signer is unavailable
- Seven-day unattended live run (days 23–30): the full pipeline anchoring every interval through the multisig at real-time pace, fed by the solar simulator and the CSV replay adapter
- Fault injections during the live run (shaded sensor, disconnected meter, night injection, implausible ramp) — each must appear in the public rejection log with its reason
- End-to-end QA
- Public write-up linking every transaction
- Short demo video

Why this matters

Seven consecutive days of timestamped transactions cannot be staged after the fact. Rather than asking reviewers to trust screenshots, the sprint ends with a live record, a page that recomputes the truth in the visitor's browser, and a rejection log that proves the gates bite.

Out of Scope

To maintain an achievable 30-day scope, the following items are explicitly excluded:

- Hardware of any kind (no meters, panels, boards or field installation)
- Token issuance, redemption or any real-world-asset claim
- Soroban smart contracts
- Mainnet deployment
- Commercial or offtake agreements
- Energy trading or energy sales of any kind
- Legal or corporate structuring
- SCADA or monitoring-platform integrations
- Marketing and community campaigns
- KYC or identity verification
- Wiring the Phase-2 batch gates (sun witness, ramp, integrity) into the spec bindings — already implemented and tested; deferred to the next Instaward

4.2 Deliverable-Aligned Budget Request

Requested Budget: USD \$3,500

The request is deliberately below the Instawards maximum. Both founders develop full time for the sprint, the gate specification and conformance suite are already shipped, and no hardware is needed. The team has planned the sprint at roughly 200 founder engineering hours; the award funds approximately 70 of them at USD 35/h plus the operating items below. The remaining hours are contributed.

Operational & Core Team (70%)

- Protocol & verification engineering
- Meter adapter development (Modbus, HTTP, CSV)
- Stellar integration (multisig, transactions, Horizon)
- Independent verifier service
- QA, conformance & adversarial testing
- Technical documentation & write-up

Technology & Operations (30%)

- Independent verifier hosting on a provider separate from the proposer
- CI and repository infrastructure

- Public PV dataset acquisition and processing
 - AI-assisted development
 - Demo & video production
 - Documentation
-

5. 30-Day Execution Plan

Week 1

Days 1–3: architecture freeze — schemas, batch definition, hash rules, adapter and verifier interfaces, transaction format

Days 4–7: Modbus adapter with register-map configs

HTTP/JSON and CSV adapters

Schema normalization

Two reference register maps

Adapter harness

Expected Output

Any of the three adapters feeds the engine and produces byte-identical verification output for the same input.

Week 2

Batch assembly and hash chain

Accepted and rejected ledgers

Evidence manifest

Batch-level conformance

Historical scale test (runs in minutes)

Expected Output

Intervals become hash-chained evidence batches, with rejections recorded and published with reasons.

Week 3

Multisig account setup

Proposer service

Independent verifier service

Third signer onboarding

Atomic anchoring transaction

Test transactions A–D

Live run starts (target day 21)

Days 21–22: integration freeze — no new features; reliability, recovery, documentation

Expected Output

The first 2-of-3 anchored batch on Stellar Testnet, with linked transactions proving a single signer cannot anchor, and the pipeline left running.

Week 4

Evidence feed

Reconciliation page

Seven-day live run (days 23–30) — architecture untouched; monitor, record, collect evidence

Fault injections into the live run

QA

Write-up and demo video

Expected Output

Seven consecutive days of unattended anchoring on Stellar Testnet, a non-empty rejection log, and a reconciliation page anyone can run.

6. Evidence of Completion

Deliverable 1

Evidence

- Public GitHub repository
- Two register-map configuration files
- Adapter harness output showing identical results across adapters

- CI run
-

Deliverable 2

Evidence

- Sample evidence batches and manifests
 - Accepted and rejected ledgers with reason codes
 - Scale-test output: an unbroken hash chain over one month of historical intervals
 - Conformance suite run (CI link)
-

Deliverable 3

Evidence

- Multisig account on stellar.expert showing the three signers and thresholds
 - Stellar Testnet transaction hashes of anchored batches
 - The four test transactions A–D, each linked
 - Independence test log: an unpublished batch the verifier refused to sign
 - Independent verifier logs
-

Deliverable 4

Evidence

- Live reconciliation page URL
 - Public evidence feed URL
 - Seven consecutive days of anchored testnet transactions (stellar.expert links)
 - Public rejection log showing the injected faults with reasons
 - Write-up linking every transaction
 - Short demonstration video
-

Evidence Verification Checklist

Deliverable	Evidence Present	Partial	Missing
Meter-Agnostic Ingestion	☐	☐	☐
Evidence Batches & Hash Chain	☐	☐	☐
Multi-Party Authorization	☐	☐	☐
Evidence Feed & Seven-Day Live Run	☐	☐	☐

7. Next-Step Alignment

Following this Instaward, QWATT will be positioned to pursue the remaining layers in separate, independently scoped grants: a second Instaward for issuance semantics on testnet (min(A,B) tokenization, floor-never-round accounting, double-mint protection, burn on redemption); a third Instaward moving the independent verifier into a Soroban contract, so the team is no longer an operator in the authorization path; and an SCF Build Award for the physical pilot (dual-metered rooftop), assurance audit, legal structure and mainnet readiness. Each stage has its own acceptance criteria and none depends on approval of the next.

On the adoption side, and outside the scope of any grant, the verified record is what the team will take to its first users: a building that shares a rooftop array and pays a small monthly fee because the record ends a dispute it already has; a first redeemer that retires QWATT for a real, checkable claim; and one renewable-attribute registry or audit firm asked what a QWATT record must look like to be accepted. Those conversations, not the token, decide the next year of engineering.
